

The Department of Computer Science

CIS4145

Business Intelligence

Group Report

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Team Member Contributions

Team Member	Dataset	Dashboard	Report	Other
Abigail Harris	Donations; Gift Shop Sales - cleaning assist; Events – checked	Commercial	Data Preparation and Cleaning – Donations; Data Integration; Data analysis – commercial; Ethics – commercial; Recommendations – commercial	V&A Research; Data Model Relationships
Belema Grend	Museum Visits	Fundraising & Donations	Introduction and Data Analysis (Fundraising and donations) Recommendations (Fundraising and donations)	V & A Research
Kate Forshaw	Events – cleaned; Feedback – broad themes	Finance	BI Requirements; Data analysis – finance; Ethics – events; Recommendations – finance	V&A Research; Data dictionary; Strategic questions
Louis Appiah-Kubi	Gift Shop Sales - cleaned	COO	Data analysis – COO; Dashboard Design (shared);	
Nichola Wright	Feedback; Gift shop sales – checking	Learning, and Young V&A; National Programmes Sentiment	Executive Summary; Data preparation and Cleaning; Thematic Analysis; Data Integration (shared); Dashboard Design (shared); Data analysis – Learning, and Young V&A; National Programmes Ethics – Feedback; Recommendations - Learning and Young V&A; National Programmes	V&A Strategy Research; Data Model Integration

Table 1: Team Member Contributions

Executive Summary

This report presents a Business Intelligence (BI) analysis and accompanying BI dashboard for the V&A museum, facilitating a reliable data-driven foundation for strategic and operational decision making across its key functional areas. By integrating and analysing data from events, visitors, donations, retail activity, and feedback, the analysis enables the museum to appraise performance, understand audience behaviour, and assess the effectiveness of its programmes in relation to its strategic plan.

Following discussion of BI requirements, expected users, project goals, and objectives, hypothesis are presented. Methodology of data cleaning is overviewed before dashboard design principles are advised. Comprehensive data analysis, findings, and insights are then presented, grouped by key V&A board roles before recommendations are offered, derived through the data analysis. Ethical and contextual considerations are also presented.

Key recommendations include scheduling value-driven programming, offers and retail initiatives across the year to stabilise income and visitor numbers. Encourage year-round giving through season-themed campaigns. Investigate website improvements to enhance the virtual visitor experience, particularly with family and student programming. Improve the consistency of visitor experience by conducting further survey and cross verification on findings of feedback thematic analysis e.g., digital experience difficulties and overcrowding.

1. Introduction

The Victoria and Albert Museum (V&A) is a world leading museum of art, design and performance, holding a collection of over 2.8 million objects spanning 5,000 years of human creativity. Since its founding in 1852, the V&A has occupied a unique position in the global cultural landscape not merely as a custodian of artefacts, but as an active institution committed to inspiring creativity and advancing public understanding of the designed world. Its mission is to enrich people's lives by promoting knowledge and enjoyment of the designed world to the widest possible audience.

1.1. Overview of the global museum sector

Museums occupy a central role in the cultural, educational, and economic life of societies around the world. According to Global Market Insights (2024), the global museum sector was valued at USD 57.2 billion in 2023 and is projected to grow at a compound annual rate of around 6% through to 2032, driven by new exhibitions, technological innovation, and rising demand for immersive cultural experiences. This growth trajectory reflects both the enduring public appetite for cultural engagement and the significant transformation underway across the sector.

Yet alongside this growth, institutions face a convergence of structural challenges: declining public and government funding, increasing competition for leisure time, rapidly shifting audience demographics, and rising operational costs. Together, these pressures have placed the long-term financial sustainability of many museums large and small under considerable strain.

At the same time, audience profiles and expectations have evolved significantly, particularly digital audiences acquired and accelerated through increased technology and COVID19 lockdown and recovery efforts (Le Dinh et al., 2024). Visitors today expect personalised, immersive, and digitally enriched experiences that extend well beyond the traditional gallery visit. Institutions that fail to adapt risk not only declining footfall, but a gradual erosion of cultural relevance. In this environment, the ability to understand audiences deeply, respond to their needs quickly, and communicate value compellingly has become a defining competitive advantage.

Research increasingly supports and advocates the role of data and business intelligence in helping cultural institutions navigate these pressures (Le Dinh et al., 2024). It has shown that organisations harnessing data and artificial intelligence make better decisions and outperform competitors on key financial metrics. In the cultural sector, institutions that invest in understanding their audiences deeply by mapping visitor journeys, identifying engagement patterns and analysing sentiment are better positioned to design effective programmes, strengthen donor relationships, and improve visitor satisfaction. The V&A is well-placed to build on these lessons and scale its operations.

1.2. The Case for Data-Driven Decision-Making at the V&A

Despite its global stature and the significant volume of activity it generates across exhibitions, events, donations, retail, and visits, the V&A currently lacks a consolidated, analytics-driven view of its performance across these dimensions. Fundraising income, visitor data, event attribution, retail revenue, and visitor sentiment are generated in significant volumes but have historically been analysed in siloes limiting the institution's ability to identify cross-cutting trends, understand the drivers of performance, or allocate resources strategically.

This gap represents both a risk and an opportunity. Without a unified intelligence framework, strategic decisions risk being made on incomplete information, perpetuating inefficiencies and missed opportunities. Conversely, a structured approach to business intelligence has the potential to unlock meaningful improvements across every dimension of the museum's

performance, from visitor experience and programme design to fundraising effectiveness and commercial revenue.

1.3. The V&A Strategic Missions

In response to these challenges, the V&A (2021) has articulated five strategic missions that together define its institutional priorities for the years ahead:

1. **Rebuild Visitor Numbers:** To restore and grow physical and virtual attendance, ensuring the museum remains accessible and compelling to diverse audiences.
2. **Transform Access & Creative Relevance:** To broaden the reach and relevance of the V&A's collections and programmes, ensuring the museum speaks to communities and audiences who have historically been underserved.
3. **Inspire Creative Confidence & Skills:** To deepen the educational and creative impact of the V&A's offer, equipping visitors particularly young people with the skills and confidence to engage with creativity in their own lives.
4. **Encourage Effectiveness, Sustainability & Inclusion:** To embed a culture of operational excellence, environmental responsibility, and inclusive practice across all aspects of the museum's work.
5. **Build Financial Resilience:** To diversify and grow the V&A's income base, spanning donations, commercial activity, and partnerships to reduce dependence on any single funding stream and ensure long-term sustainability.

1.4. Scope and Objective of Report

This report presents a business intelligence-driven analysis across five key operational areas: donations and fundraising, exhibitions and events, gift shop sales, physical and virtual visits, and visitor feedback. Drawing on data from 2023 and 2024, the analysis is designed to support executive decision-making by identifying the key drivers of visitor engagement, evaluating revenue performance across income streams, and highlighting strategic opportunities to strengthen the visitor experience and the museum's financial position. Each area of analysis is considered in the context of the five strategic missions, ensuring that findings and recommendations are anchored to the V&A's institutional priorities.

2. Business Intelligence (BI) Requirements

2.1. Purpose

The purpose of the BI project is to provide the V&A with a reliable, data-driven foundation for strategic and operational decision-making across its key functional areas. By integrating and analysing data from events, visitors, donations, retail activity, and feedback, the system enables the museum to monitor performance, understand audience behaviour, and assess the effectiveness of its programmes in relation to its strategic plan. The BI dashboards are designed to give senior leaders and departmental teams clear, actionable insights that support financial sustainability, enhance visitor engagement, strengthen donor relationships, and optimise operational efficiency. Ultimately, the BI solution ensures that decisions are informed by accurate, timely, and meaningful information, enabling the V&A to deliver public value while upholding its organisational values.

2.2. V&A Strategic Context

The BI system is designed in alignment with the V&A's strategic plan, which centres on rebuilding visitor numbers, broadening connections with diverse audiences, inspiring creative confidence, transforming access and relevance, and strengthening financial resilience (V&A, 2021). These mission pillars shape the strategic questions explored across the five dashboards and ensure that the analyses support organisational priorities rather than isolated metrics. By grounding the BI requirements in the museum's strategic direction, the system provides insights that are meaningful, actionable, and directly connected to the V&A's long-term goals.

2.3. Users

- Director, Deputy Director & Chief Operating Officer (COO) – optimises internal operations and ensures a seamless visitor experience (V&A, 2026).
- Director of Learning, National Programmes & Young – leads educational programming, audience engagement, and accessibility initiatives (V&A, 2026).
- Director of Finance & Resources – oversees financial sustainability and manages the balance of income streams (V&A, 2026).
- Commercial Director – directs monetisation and marketing strategies to maximise returns while maintaining visitor experience quality (V&A, 2026).
- Director of Development – manages fundraising strategy and cultivates donor relationships (V&A, 2026).

2.4. Goals

1. Identify attendance patterns, satisfaction levels, and differences between physical and virtual engagement to highlight potential inefficiencies, understand visitor demand, and support operational decisions related to staffing, scheduling, resource allocation, and programme continuity.
2. Provide clear visibility of visitor satisfaction trends and event-specific engagement to support improvements in the visitor experience and help identify the drivers of positive and negative feedback.
3. Support financial planning by analysing revenue diversification, the contribution of different income streams, and the relationship between attendance and revenue to inform long-term financial sustainability.
4. Inform merchandising and pricing strategies by analysing gift shop performance, average spend per visitor, product category trends, and the relationship between event popularity and retail revenue to strengthen commercial outcomes.
5. Support fundraising strategy by tracking donation trends, donor engagement patterns, seasonal fluctuations, and the relationship between event attendance and donation behaviour to enable more targeted and effective fundraising initiatives.

2.5. Objectives

- Integrate and clean each dataset to ensure accuracy, consistency, and analytical readiness, enabling reliable insights across all dashboards.
- Answer clearly defined strategic questions that reflect the V&A's mission pillars and organisational priorities.

- Translate the needs of primary users into targeted analyses and metrics that directly support their decision-making responsibilities.
- Present clear, intuitive visualisations that address strategic questions and enable users to interpret performance quickly and confidently.
- Deliver interconnected dashboards that provide a unified view of organisational performance while serving the distinct needs of each user group.

2.6. Hypotheses

- There will be a significant difference in revenue generated across the museum's income streams, and revenue levels will vary across different periods of the year.
- Diversified programming in 2024, towards student and family-targeted exhibitions and events (over adult programming), has increased museum sentiment, engagement and visitor numbers.
- Commercial revenue is primarily driven by retail activity rather than ticket sales, indicating the importance of secondary spending in overall income generation.
- Donor engagements at the V&A are significantly influenced by event programming, donor identity and seasons.
- Enhancing data capture and integration across systems will improve decision-making speed and lead to more effective future strategies.

2.7. Value of the V&A

The BI project must reflect and support the V&A's core values of equity, creativity, collaboration, sustainability, and generosity (V&A, 2021). This requires presenting insights in a way that promotes equitable understanding of audience engagement, avoids bias, and supports inclusive decision-making across all visitor groups. Creativity is enabled through visualisations that encourage exploration and new ways of interpreting performance. Collaboration is strengthened by providing a consistent, shared evidence base across departments, allowing Finance, Commercial, Young V&A, Donations, and COO teams to work from aligned insights. Sustainability is reinforced by enabling data-driven decisions that optimise resource use, operational efficiency, and long-term financial resilience. Generosity is embodied by making insights transparent, accessible, and supportive of the museum's mission to serve the public good. Together, these values shape how the BI solution is designed, interpreted, and used across the organisation.

3. Data Preparation and Cleaning

Dataset cleaning was split equally between the project team, with each dataset checked by a second colleague. Cleaning encourages data quality, defined by Miller (2017) as accuracy, consistency, completeness, relevance, and appropriateness. Fundamentally, this process encourages for business intelligence purposes, accurate analysis and reporting whilst easing dataset integration challenges. Insights derived from cleaned data have increased validity, evokes heightened trust of BI stakeholders, and encourages confidence to use findings to make data driven decisions. In contrast poor quality datasets can distort patterns and cause harm by misleading information (Sharman, 2022).

Across each dataset, columns containing missing data were assessed and where appropriate rows removed or values imputed (e.g., quantity sold derived from total sales value ÷ unit price). In select scenarios missing values were maintained, such as with gender and age fields on the Feedback dataset. Survey design may have facilitated these as optional equality fields, common on surveys; whilst removal of rows for unknown gender would have removed 500+ otherwise valid comments. Similarly, exclusion of unknown visitor type would have distorted overall gift shop sales by £17,167 and donation totals by £11,280, misaligning data analysis with other company metrics.

Dataset cleaning methodology accompanies each individual dataset.

A comprehensive data dictionary covering all datasets is provided in [Appendix A](#) detailing dataset fields, purpose, format and valid entries.

3.1. Feedback Dataset Thematic Analysis

The “Feedback” dataset contained written comments provided from museum visitors (both physical and virtual) and covered all events and museum venues within the two-year period. Following dataset cleaning 3138 comments were available for thematic analysis.

To facilitate sentiment analysis individual comments were first reviewed and assigned an overarching sentiment category (positive, mixed or negative). Next, thematic analysis occurred at two levels – broad and narrow to gauge shared trends for follow up and decision-making purposes. Praised aspects within positive and mixed sentiment were brought together in “positive broad theme” and “positive narrow theme”. Similarly, criticised themes from negative and mixed responses were gathered in “negative broad theme” and “negative narrow theme”. Please see [Appendix B](#) for theme categorisations. The V&A BI dashboard presents the top positive and negative sentiment themes interactively to allow discovery of patterns.

3.2. Donations Dataset Preparation and Cleaning

The “Donations” dataset contained records of financial contributions, including donation values, associated events and donor attributes. To ensure consistency across datasets, categorical fields such as “Event Name” were cleaned and standardised, with missing and inconsistent entries consolidated under a single “Unknown” category. This improved compatibility with the Exhibitions and Events dataset and enabled accurate relationship mapping. In cases where donations were not linked to a specific event, classification as “Unknown” did not adversely impact analysis, as donation value and overall contribution patterns remained the primary focus of financial evaluation; this also reflects real world scenarios in which donations are not always tied to specific events. Fields such as donation amounts were reviewed and formatted to support reliable aggregation and comparison. A high value outlier (£1,000,000) was identified within the dataset and retained, as it represents a legitimate transaction and contributes to an accurate reflection of overall donation patterns.

3.3. Data integration

To enable cross-dataset analysis, the five museum datasets were integrated within Excel using PowerPivot, where relationships between tables were established within the Data Model. A snowflake schema was constructed in which the Exhibitions and Events dataset functioned as the central dimension table, as it contained the primary records linking Events across the remaining datasets. The Museum Visits, Gift Shop Sales, Donations and Feedback datasets were modelled as fact tables, with each contributing attributes that enriched analytical depth. This approach aligns with established data-integration principles, where multi-source analytical systems require structured schemas, conformed dimensions, and harmonised keys to support consistent querying and insight generation (Koukaras, 2025).

To support relational integrity, several additional helper tables were created, including a Date table, a Visitor Type table, a Category table and a Museum table. These tables were used to resolve inconsistencies between datasets and enable stable relationships within the Data Model, reducing ambiguity and supporting accurate cross-dataset analysis. The need for such harmonisation is well-documented in heterogeneous data-integration research, which emphasises schema matching, entity resolution, and semantic alignment as essential steps when combining structurally diverse datasets (Koukaras, 2025; Putrama and Martinek, 2024).

Further data cleaning and standardisation were required to resolve inconsistencies across the different datasets. For example, the Donations dataset categorised “virtual” visitors as “online” visitors, whilst the Museum Visits dataset used “virtual”. These were standardised into a single controlled vocabulary to avoid fragmented categories in the model. Similarly, column names differed across datasets, for instance, “region” in the Museum Visits dataset correlated to “museum” in the Donations dataset, requiring renaming and alignment to ensure a consistent relationship. Such challenges are typical in heterogeneous data environments, where semantic inconsistencies and structural heterogeneity must be addressed before integration can occur effectively (Putrama and Martinek, 2024).

Once the datasets were cleaned and the relationships were established, the integrated model enabled exploratory data analysis. This allowed measures such as visitor behaviour, donation patterns, event performance and sales insights to be analysed holistically. The analytical value of this approach reflects broader findings in literature: integrated multi-source datasets support richer insights, improved decision-making, and more robust analytical workflows compared to siloed data structures (Koukaras, 2025).

The resulting relational structure is shown in the Power Pivot data model diagram, which visually represents the snowflake schema and table relationships (see Figure 1).

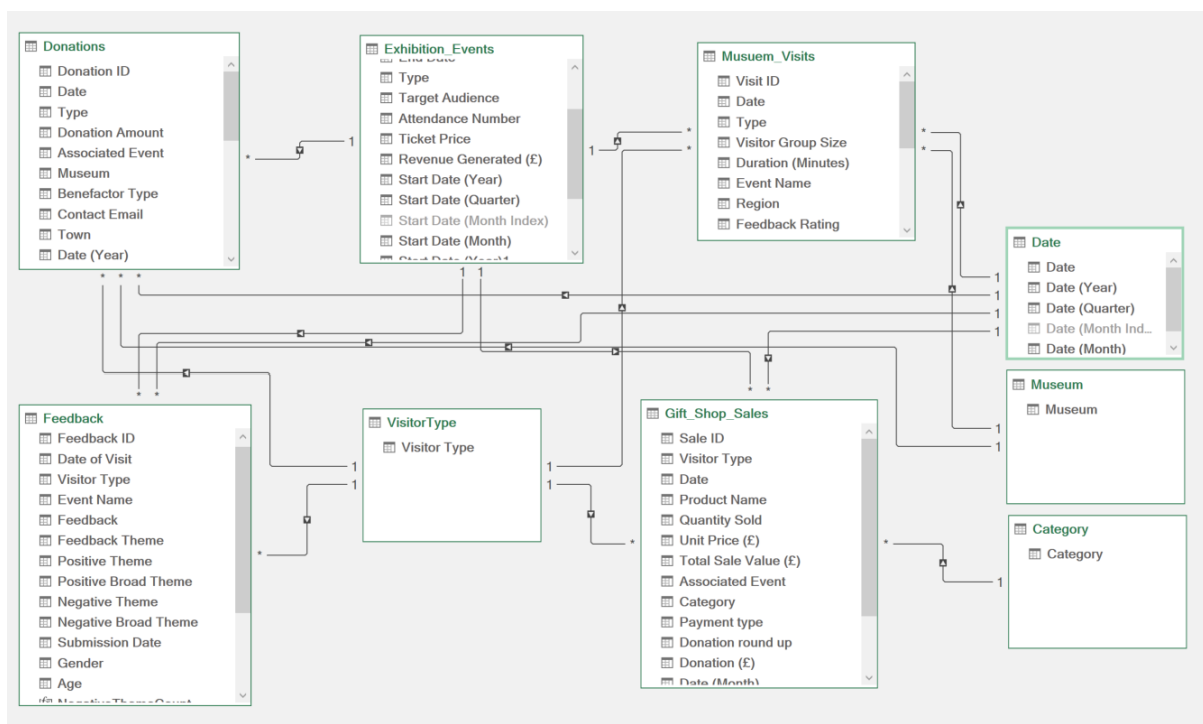


Figure 1: Data Model Diagram

4. Dashboard Design

The V&A strategy dashboard has been developed in Microsoft Excel, a popular, familiar, and cost-effective tool for data hosting, transformation, dataset connectivity (via Power Query), analysis and visualisation (Chandra and Dwivedi, 2022). Graphical presentation via pivot charts facilitates option for interactive visuals enabling pattern discovery; whilst underlying pivot tables offer options to slice into datasets (within purpose, user and data privacy bounds).

Structural design of the business intelligence dashboard has focussed on V&A executive board roles with each tab hosting visuals which address key thematic questions aligned to two or more mission areas of the 2021-2025 strategic plan (V&A, 2021). Strategic questions considered during analytical exploration are presented in [Appendix C](#).

The dashboard therefore aims, as promoted in academic literature (Alzghoul et al., 2024; Mohiuddin, Mamun and Zamil, 2025), to deliver valuable insights to support performance monitoring and data-driven decision making; whilst promoting strategic success and impact through executive alignment, collaborative planning and recognition of interconnected activities.

Achieving the right balance, volume and style of visualisations on the dashboard has been an important design decision and one that researchers have found to impact perceived and realised dashboard quality (Burnay, Lega and Bouraga, 2024).

The team has sought to integrate both static and interactive charts within the overall design to balance data storytelling with utility (i.e. reconfigurable charts for pattern exploration), thus encouraging insight and decision-making potential from the tool.

The design of the dashboard drew inspiration from principles of Gestalt psychology, which explain how users intuitively organise visual information. Principles such as similarity and proximity were applied to group related metrics, while continuity supported a logical flow aligned with user reading patterns. Figure-ground differentiation was used to prioritise key insights, and connectedness reinforced relationships between data points. Closure enabled simplified visuals without loss of meaning, while common fate was reflected in interactive elements that updated multiple visuals simultaneously. Finally, Prägnanz (simplicity) underpinned a clean, uncluttered design, enhancing usability, reducing cognitive load, and supporting effective decision-making. For example, metrics that were related were grouped, consistent visual styles were used, and a logical layout guides user attention. Gestalt principles offer faster information processing, improved usability, and more accurate decision-making. In an executive context, where rapid understanding of complex data is required, structuring information clearly and easily perceivable enhances clarity, engagement, and the overall effectiveness of the dashboard as a strategic decision support tool (Aliyev, 2024).

The volume of charts on each tab, as well as informational and non-informational content also required careful consideration. Overloading a dashboard can dilute the impact of key findings, confuse, and reduce ability to make decisions, and risk tool underutilisation due to users perceiving or experiencing cognitive overload; thus, impacting assessment of usefulness, ease of use, and enjoyment (Burnay, Lega and Bouraga, 2024). In contrast, underloading a dashboard may deliver too few insights to answer key questions, enable decision making, or encourage action or repeat use (Burnay, Lega and Bouraga, 2024).

Interactive visualisations may increase cognitive load of the dashboard; however these informational aspects have been included as they can enhance engagement levels of both primary and secondary users with a BI tool (including front line staff if presented), increase strategy understanding and legibility, and galvanise shared action or follow up on discerned patterns (Azad and Zablith, 2021).

5. Data Analysis, Findings and Insights

5.1. Chief Operating Officer

Mission 1: The attendance analysis shows that three flagship exhibitions (*Japan: Myths to Manga*, *Jameel Prize: Moving Images*, and *Tropical Modernism*) collectively attracted over 16,000 visitors, accounting for nearly all exhibition-based footfall. This concentration proves the museum’s strong ability to draw audiences through high-profile cultural programming but also highlights a dependency on a small number of major exhibitions to drive visitor recovery (see figure 2).

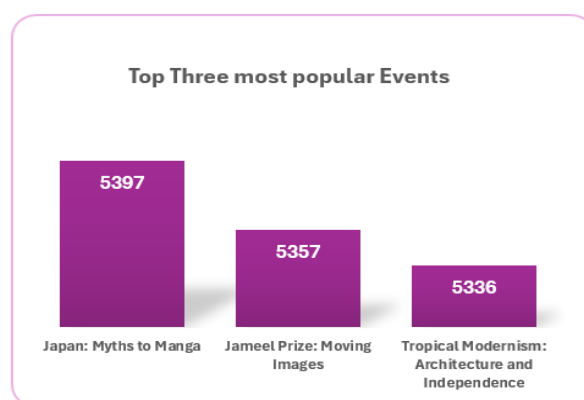


Figure 2: Top 3 events by footfall

Mission 2: Visitor group size data indicates balanced engagement across all V&A sites, with South Kensington, Dundee, East Storehouse, and Young V&A each attracting between 3,200–3,700 visitors in 2024. The comparable scale of online engagement further suggests that digital access is becoming an equally significant channel. This distribution reflects broad geographic reach but also signals opportunities to deepen engagement among underserved communities (see figure 3).

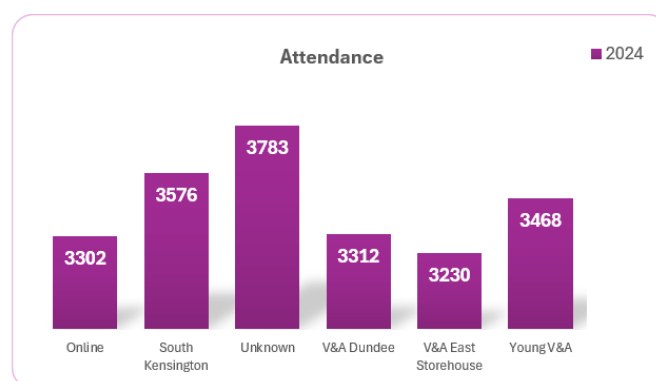


Figure 3: Attendance by footfall

Mission 3: Audience composition analysis reveals that students represent the largest visitor segment, accounting for over 70% of total attendance. This indicates that the museum’s educational and creative programming is highly effective at engaging younger audiences.

However, the comparatively lower participation from adults and families suggests untapped potential to broaden creative learning pathways across age groups (see figure 4).

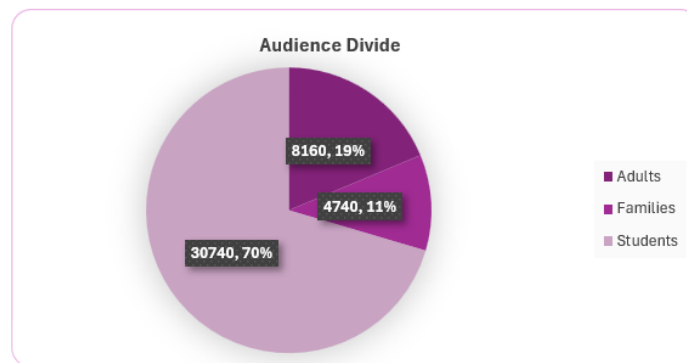


Figure 4: Attendance Groups

Mission 4: Feedback theme analysis shows that major exhibitions generate the highest levels of positive sentiment, while workshops and smaller events receive disproportionately high negative feedback. This pattern suggests that while large-scale exhibitions deliver strong visitor experiences, operational or delivery challenges within smaller programmes may be limiting inclusivity and overall satisfaction. Addressing these inconsistencies would strengthen the museum’s commitment to effective and equitable practice (see figure 5).

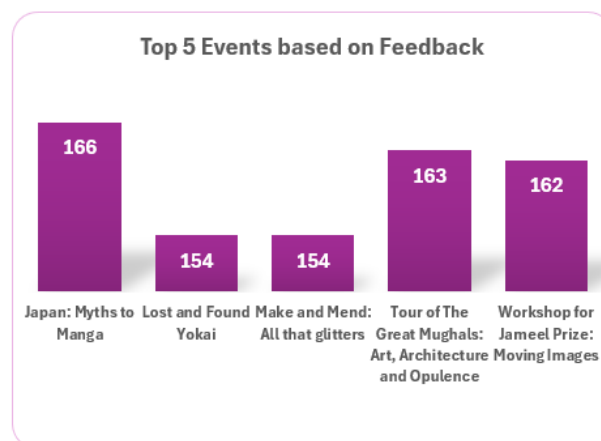


Figure 5: Top 5 events by feedback

Mission 5: The income stream analysis reveals that ticket Donation revenue (£1,602,388.58) significantly exceeds both ticket (£262,122.00) and retail income (£600,942.26), indicating a strong reliance on donations as the primary revenue driver in 2024 as compared to 2023 where the primary driver was ticket sales (694,457.59). The total Revenue at face value looks positive due to an increase in revenue but diving deeper reveals that an outlier of (£1,000,000) was donated which boosted the overall revenue to an overall average of a 42% boost in revenue. However, removing the outlier actually points to an overall dip of approximately 16% overall in revenue. This means there has to be a bigger push into ticket sales and retail sales, and this can be achieved through an increase in overall attendance, as well as addressing any negative feedback that has yet to be addressed to improve.(see figure 6).

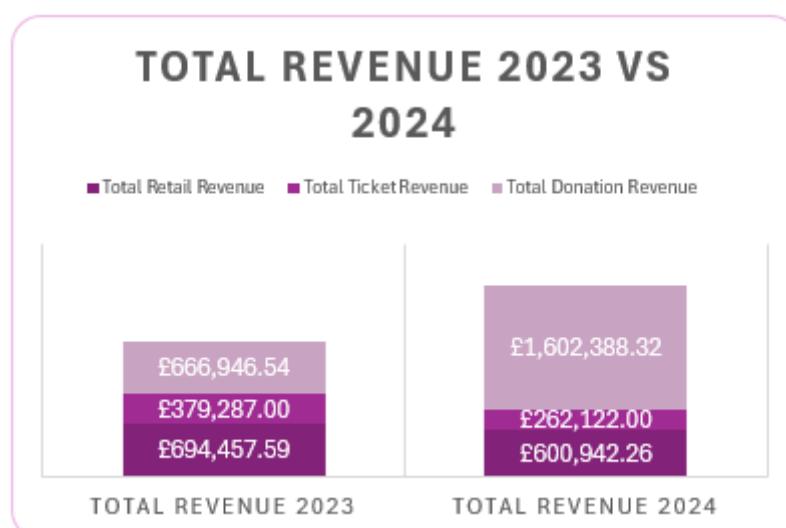


Figure 6: Total Revenue

5.2. Fundraising & Donations Dashboard

Mission 5: Fundraising has grown dramatically year-on-year. Total donations more than doubled from £612,313 in 2023 to £1,657,021 in 2024, a 170% increase representing a significant milestone in the museum's financial recovery. This level of growth signals increasing confidence from the donor community in the V&A's programmes, leadership, and long-term vision. However, sustaining this trajectory will require deliberate stewardship of existing donors and a structured approach to identifying and cultivating new giving relationships.

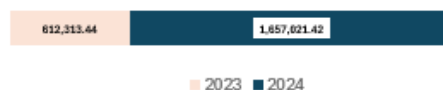


Figure 7: Donations by year

Mission 5: There was an increase in donors to the museum from 2023, however this is not at par with donations received. The number of donors rose by 14% (from 1,382,128 to 1,574,278), while donations grew by 170%. This disparity suggests that a small number of high-value donors are likely driving the bulk of revenue growth, creating a concentration risk that could leave the museum financially exposed should any of those relationships change. Building a broader and more balanced donor base across all giving levels should therefore be treated as a strategic priority.

Mission 4: Over half of all donated income is completely unattributed. The single largest donor segment is recorded as Unknown at £1,000,000 exceeding the combined totals of Family (£601,615), Individual (£592,175), and Corporation (£75,543) donors. This represents a significant operational gap, as the museum is currently unable to identify, thank, or meaningfully engage with the source of more than half its donated income. Without addressing this data quality issue, the V&A's ability to build lasting donor relationships and demonstrate accountability to its funders will remain fundamentally limited.

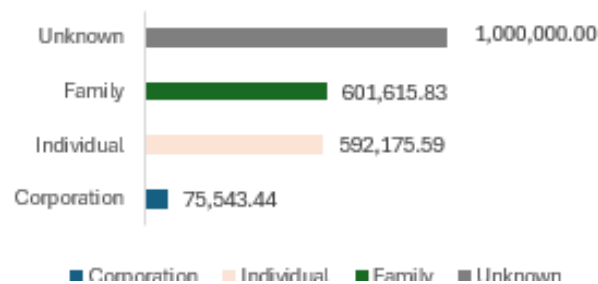


Figure 8: Donor type

Mission 5: The donation calendar is heavily skewed toward year-end. The time-series chart shows that donations remain minimal from February through October, with a dramatic surge concentrated in November and December. This pattern creates significant financial vulnerability, as the museum's fundraising performance is effectively dependent on a narrow two-month window each year. A more distributed giving calendar would not only reduce this risk but also provide more predictable income to support year-round programme delivery and operational planning.

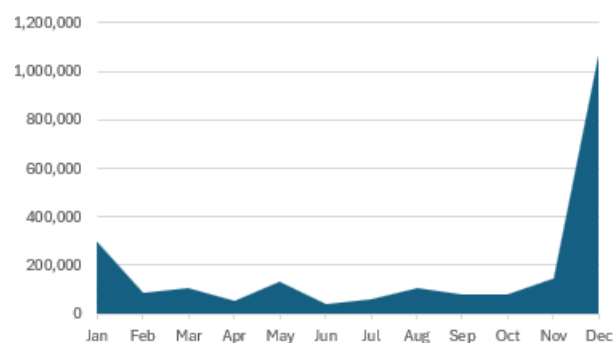


Figure 9: Donation period

Mission 4: Event-linked fundraising attribution is largely missing. While the Special Viewing of Cartier is the top identified event at £124,999, a further £1,000,000 in donations has no event association whatsoever, meaning approximately 80% of income cannot be linked to any specific programme or exhibition. This absence of attribution makes it impossible for the museum to evaluate which events and exhibitions are most effective at driving philanthropic support. Closing this gap is essential if the V&A is to make evidence-based decisions about how it invests in its programme offer going forward.

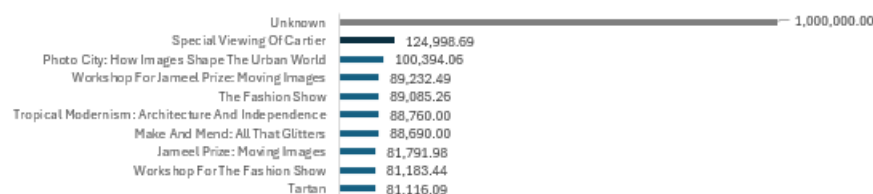


Figure 10: Donations by event

Mission 1 & 2: Visitor numbers are growing, but nearly a third of visitors are leaving with a negative impression. Visitor numbers grew by 10% from 5,049 in 2023 to 5,530 in 2024, which is an encouraging sign of recovery and growing audience engagement. However, of the 3,138

visitors who shared sentiment, 29% reported a negative experience and a further 29% were mixed meaning only 42% left with a positive impression. Rebuilding visitor numbers sustainably requires not just attracting more people through the doors but ensuring that the experience they receive is consistently positive, memorable, and reflective of the V&A's world-class reputation.

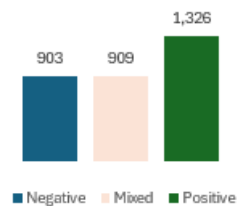


Figure 11: Visitor Reviews

5.3. Commercial Dashboard

Mission 1: The bar chart presents total visit counts across the leading exhibitions, calculated using the number of recorded visit instances per exhibition. Visit volumes are tightly clustered, with all major exhibitions receiving a similar number of visits, and “Lost and Found Yokai” recording the highest count. This implies that the exhibition programme is effective in maintaining consistent levels of visitor activity across multiple shows, supporting the objective of rebuilding and sustaining engagement across audiences (see figure 12).

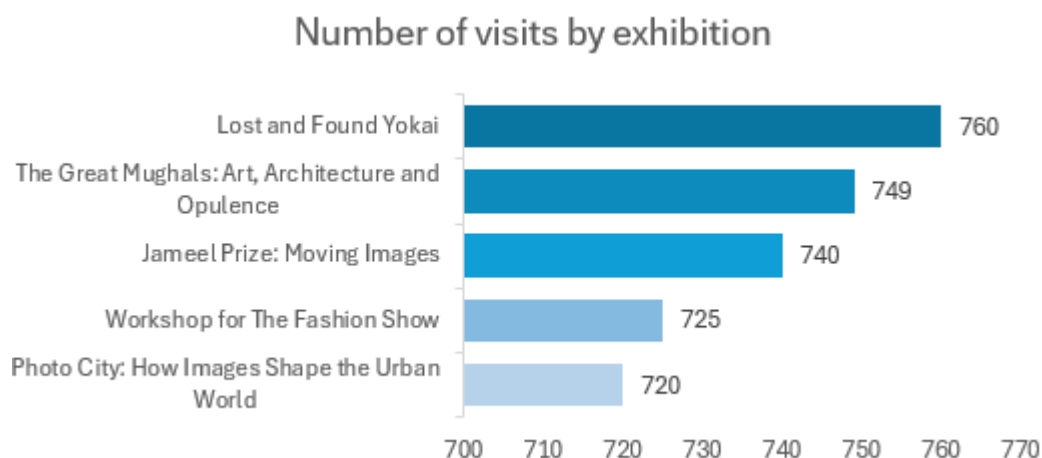


Figure 12: Number of visits by exhibition

Mission 2: The sentiment chart shows that 42% of feedback is classed as positive, with the remainder reflecting neutral or negative responses. This confirms that visitor experience is mixed rather than consistently favourable. The variability in sentiment indicates some exhibitions fail to engage audiences or deliver impact, suggesting that there is room to improve engagement, interpretation and overall consistency.

Mission 3: Average visit duration across exhibitions is highly consistent, with all major exhibitions recording average durations of approximately 121 minutes. This shows that visitors spend a similar amount of time engaging with content regardless of the specific exhibition.

Consistent time spent on site hints that exhibitions successfully maintain visitor attention and support immersive experiences.

Mission 2 & 3: The comparison of average visitor duration by event type shows that Installations (121.2 minutes) and Workshops (121 minutes) record slightly higher durations than Exhibitions (119 minutes). This demonstrates that participatory and immersive events are associated with marginally longer engagement. Workshops in particular represent interactive learning environments that directly contribute to skill development and creative confidence (see figure 13).

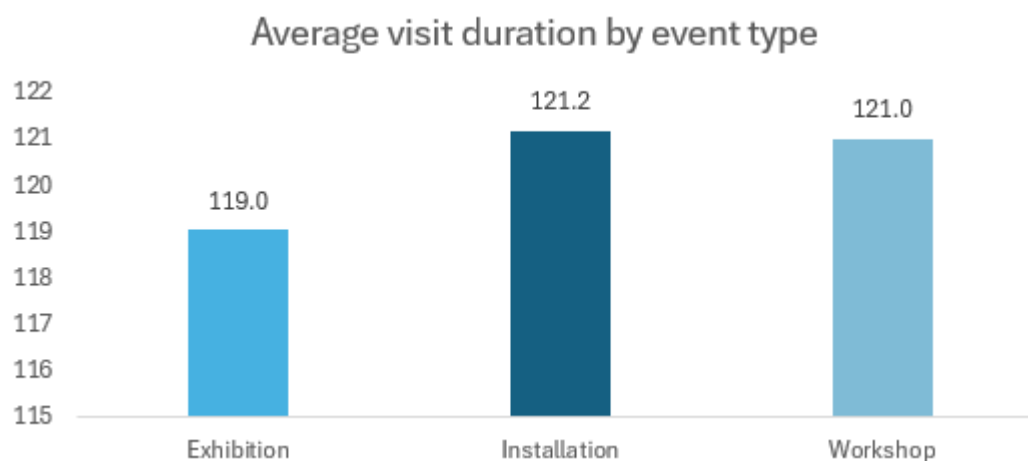


Figure 13: Average visit duration by event type

Mission 1 & 4: The monthly trend shows clear variation in visit frequency across the year, with a pronounced peak in January followed by fluctuating levels in subsequent months. This demonstrates that visitor demand is influenced by factors such as seasonality or event scheduling. Identifying peak and low periods supports strategic planning by enabling targeted programming and marketing to attract audiences during quieter periods (see figure 14).



Figure 14: Monthly visit patterns

Mission 5: The treemap highlights a clear concentration of retail revenue within a small number of product categories, with Homeware bringing in £349k, Stationery £276k, and Accessories £228k thus generating the majority of income. This confirms that commercial performance is driven by a limited number of high-performing categories. Identifying these key revenue drivers supports financial resilience by informing product strategy and focusing resources on the most profitable areas of retail activity (see figure 15).



Figure 15: Product category performance

Mission 5: The revenue split between retail and ticket sales shows that retail sales account for 67% of combined revenue, whilst ticket sales contribute 33%. This confirms that retail activity represents the dominant source of commercial income within the dataset. The findings highlight the importance of secondary spending in supporting overall financial performance and demonstrate that retail operations play a critical role in sustaining revenue beyond admissions.

Mission 4 & 5: The analysis of average spend per visitor type shows that Virtual (£153) and Physical (£150) visitors generate almost identical revenue per visit. This demonstrates that both digital and in-person channels are equally effective in generating value. The consistency in spend suggests that digital engagement is successfully integrated alongside physical experiences, supporting inclusive access to the V&A's offerings (see figure 16).

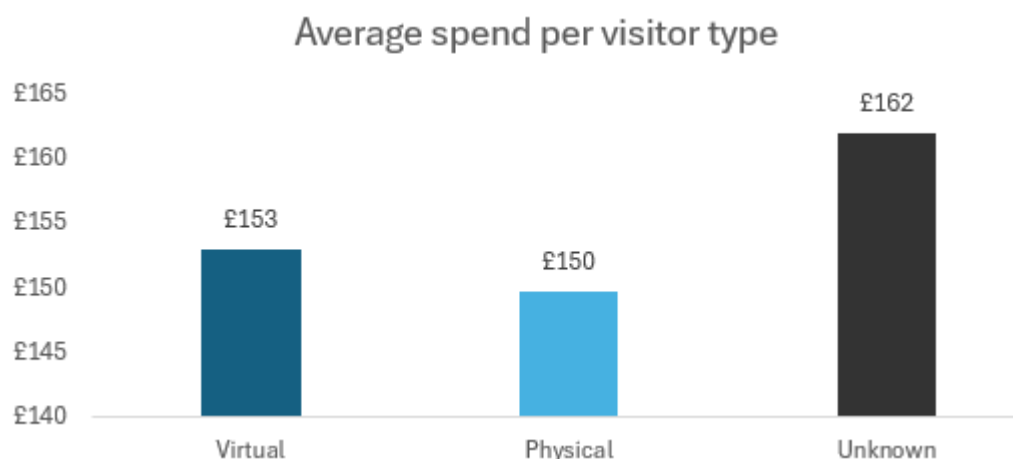


Figure 16: Average spend by visitor type

5.4. Finance dashboard

Mission 5: The income stream analysis shows that donations generate the highest revenue, significantly exceeding both retail and ticket income. This indicates strong philanthropic support but also highlights a potential concentration risk, as the organisation is heavily reliant on a single income source. Retail revenue performs well as a secondary stream, while ticket revenue contributes the least (see figure 17).

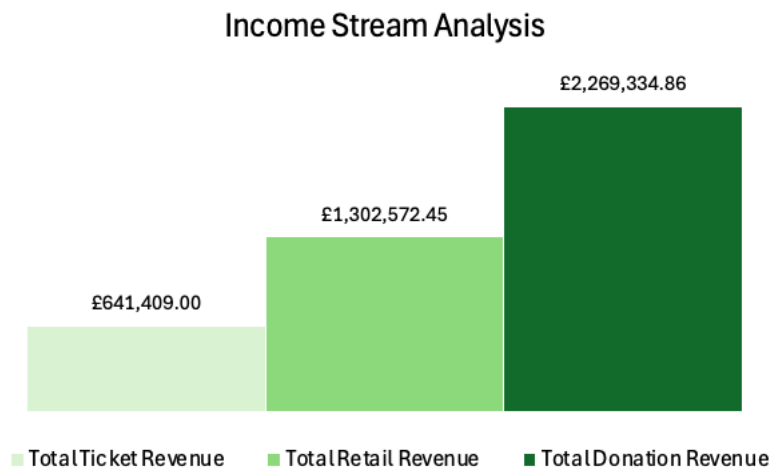


Figure 17: Income stream analysis

Mission 5: The quarterly revenue chart demonstrates that event income peaks sharply in Q1, with a noticeable decline across Q2, Q3, and Q4. This pattern signifies a strong seasonal dependency, where early-year programming drives most event-based revenue. Further, the month-by-month analysis shows that both retail and donation revenue peak in January, while donation experiences a significant surge in December. This suggests that financial performance is reliant on visitor activity and donor engagement in the winter (see figure 18 & 19).

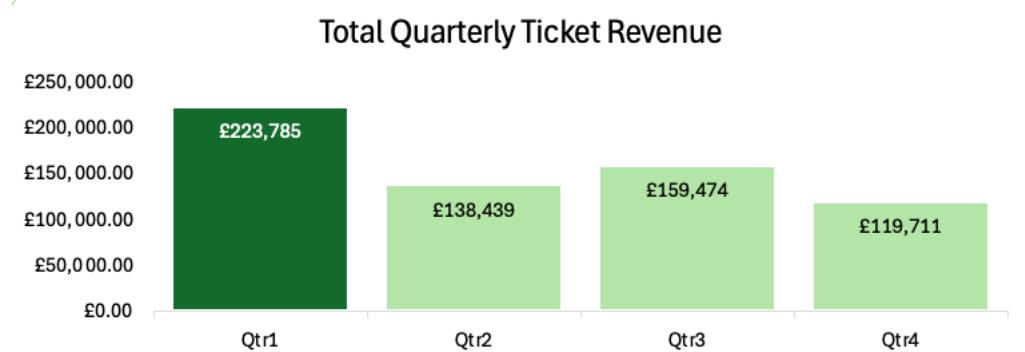


Figure 18: Total quarterly ticket revenue

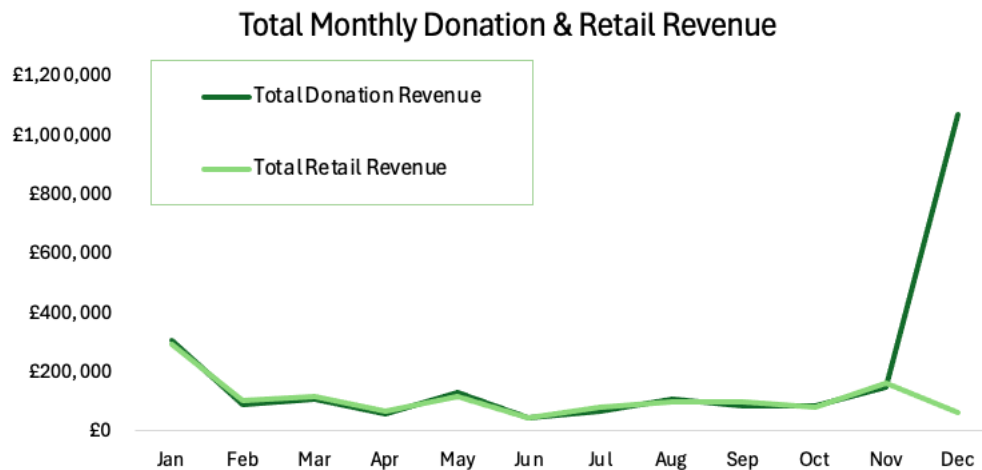


Figure 19: Total monthly donation & retail revenue

Mission 1 & 3: The audience-based revenue breakdown exhibits that student-targeted events generate the most ticket revenue, accounting for over two-thirds of income. Adults and families contribute much smaller proportions. With the museum’s strongest financial returns coming from student engagement, it infers that pricing, programming, and marketing strategies aligned with this demographic are effective. Moreover, the lower contributions from adults and families highlight opportunities to diversify revenue (see figure 20).

Ticket Revenue by Target Audience

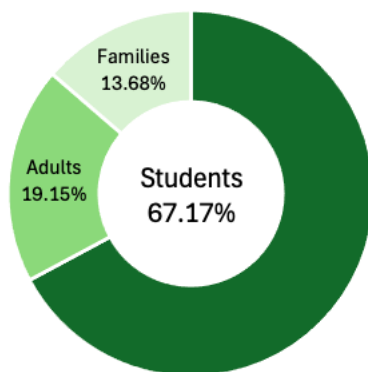


Figure 20: Ticket revenue by target audience

Mission 1 & 5: The event revenue ranking establishes that the Japan: Myths to Manga exhibition generates the highest income, closely followed by Tropical Modernism, NAOMI In Fashion, Cartier, and Photo City. This demonstrates that certain exhibitions consistently outperform others, providing a clear indication of what themes and formats drive commercial success. It also inadvertently identifies lower-performing events where adjustments may be needed (see figure 21).

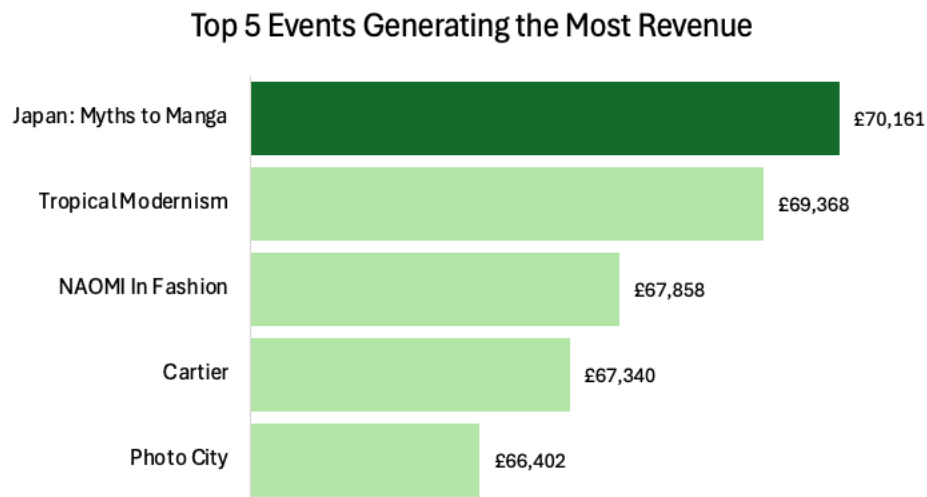


Figure 21: Top revenue generating events

Mission 5: The comparison of donation behaviour between physical and virtual visitors presents that round-up contributions outweigh direct donations for physical visitors, whereas virtual visitors give more direct donations than round-ups. This implies that micro-donation mechanisms embedded in retail transactions are particularly effective for on-site audiences, while virtual visitors are more responsive to intentional, standalone donation prompts (see figure 22).

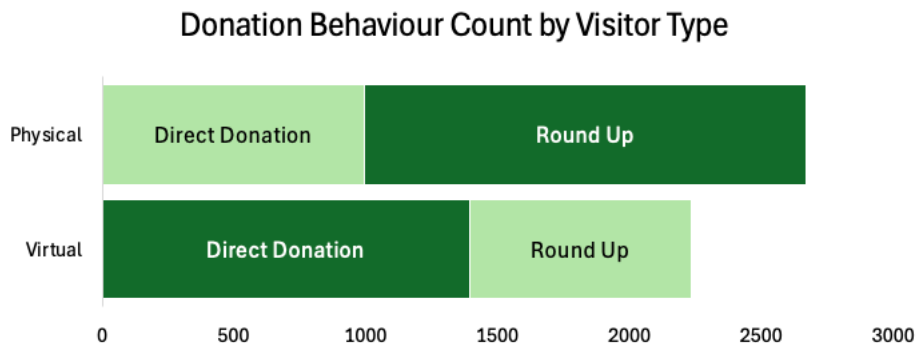


Figure 22: Donation behaviour count by visitor type

Mission 4 & 5: The scatter plot comparing revenue and attendance reveals a weak negative relationship across ticket, retail, and donation income, indicating that higher revenue does not consistently correspond with higher visitor numbers. Instead, events generating the most income tend to attract moderate rather than peak attendance levels, suggesting that financial performance is influenced more by factors such as pricing, event theme, or donor behaviour than by volume of visitors alone. Further, ticket revenue will only account for physical visitors, indicating the negative relationship with all types of attendees (see figure 23).

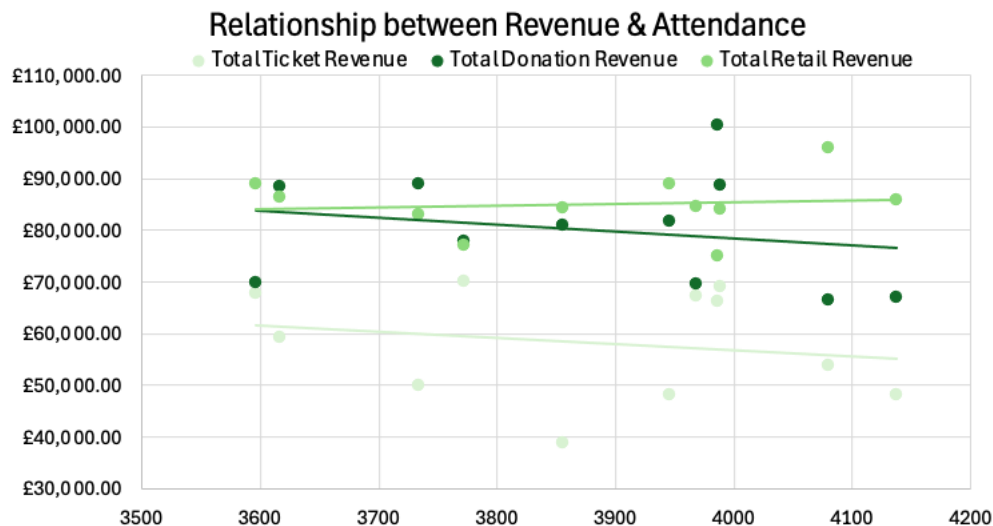


Figure 23: Relationship between revenue and attendance

5.5. Learning National Programmes and Young People dashboard

Mission 1 and 3: Exhibition programming shifted significantly between 2023 and 2024, anticipated to link with attracting, inspiring and diversifying the V&A audience. New family programming saw three exhibitions scheduled to rebuild audiences and encourage creative confidence in generation A: this reduced adult-focused programming by 75% year on year. Student-targeted events made up 40% of the 2024 event schedule (see figure 24).

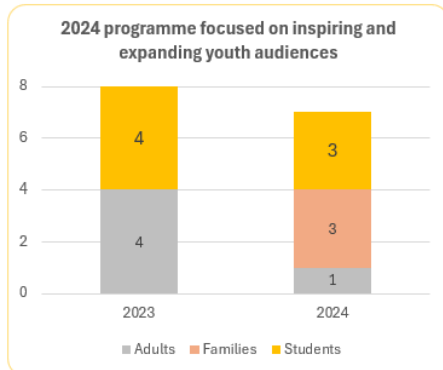


Figure 24: Shift in audience focus 2023-2024

Mission 1 and 3: Analysis of attendance across exhibitions held between 2023 and 2024, informs that student-focused exhibitions attracted the highest average attendance per event (4391); 9% more than adult-focused exhibitions which were programmed in 2023 (see figure 25). Family-targeted events in 2024 averaged 2360 attendees per exhibition; 41% lower than adult programmed events from 2023. However, continued programming and review of promotion may build future audience numbers.

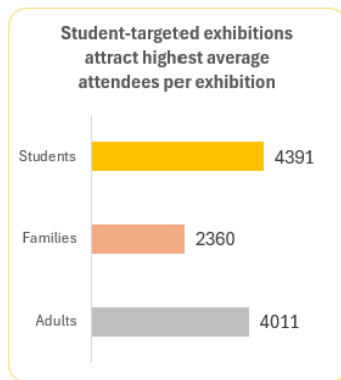


Figure 25: Average attendance by audience type

Mission 3: In considering the Young V&A’s mission to grow audiences and inspire joy, average feedback scores over time were reviewed. Scores from in person and online visitors appear to indicate middling satisfaction over time with Young V&A (see figure 26). Virtual visitors have greater fluctuation in average feedback score, albeit their cohort is smaller and thus impacted greater by swings in sentiment. Onsite visitors in 2024 appear to be slightly less satisfied than 2023, particularly between May and September when the “Tartan” and “Jameel Prize: Moving Images” exhibition were programmed. Projection of Young V&A feedback scores for October to December 2024 indicate continued decline (based on linear trend analysis).

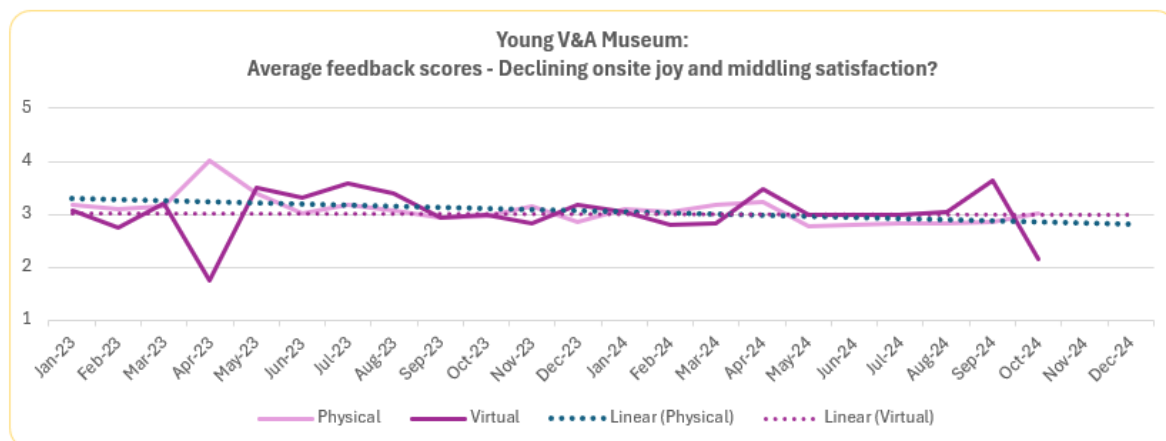


Figure 26: Average feedback scores over time

Mission 2 and 3: In considering which events resonate with virtual audiences, the highest average feedback scores correlated with workshops (3.09) indicating greater audience engagement with creative confidence programming (see figure 27). These specialist events with limited attendees achieved +0.09 on average feedback scores on exhibitions (3.01). Tour installations, held once in 2023, did not resonate with audience members who provided feedback, attaining the lowest average score (2.89). The quantity of written feedback received for student-targeted exhibitions was equivalent to 1 in 20 visitors providing feedback (i.e., 5.4%, if each submission is considered the opinion of 1 visitor rather than a group). In analysing opinion data, caution must therefore occur as feedback may not be representative of sentiment across all visitors, or specific cohorts. Recommendation to triangulate findings with other sources is thus made (e.g., social media, Trip Advisor etc.).

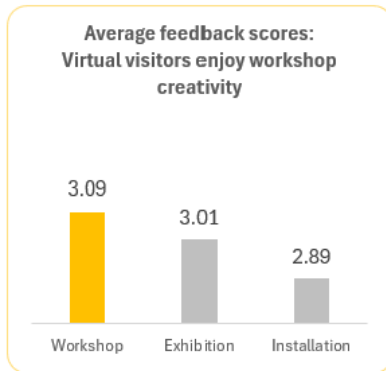


Figure 27: Average feedback scores by event type

Mission 3: Thematic analysis, conducted on the Feedback dataset to explore young peoples experience and connection with student-targeted exhibitions (i.e., from written feedback), indicated that mixed feedback (containing both positive and negative themes) was most prevalent from under 18's and 18-24 years olds - tied to virtual interactions. Interestingly, the % of respondents providing positive-only comments grew with age for virtual based student-targeted events, indicating that adult audiences enjoyed this content; and/or the positioning of content requires adjustment to meet generation Z and A expectations (see figure 28).

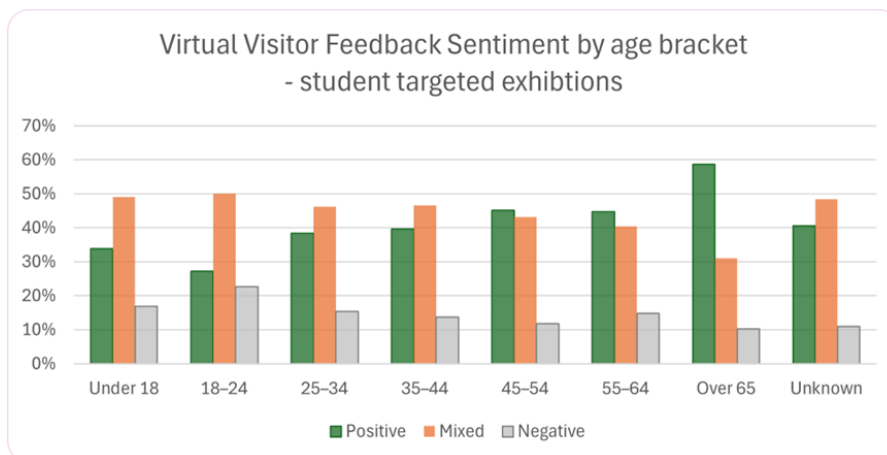


Figure 28: Virtual visitor feedback by age

Opinion analysis also identified that under 18s visiting student targeted exhibitions in person provided more negative sentiment (40%, compared to 26% mixed and 34% positive). In contrast, 18-24-year-olds expressed greater positive sentiment with 48% of comments being solely positive, 20% mixed and 32% negative (see figure 29).

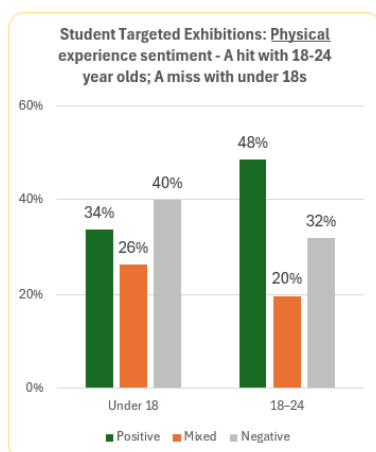


Figure 29: Audience sentiment comparison

The top 3 positive themes from positive and mixed comments for student-targeted exhibitions were exhibition quality (19.9%), enjoyment with interactive exhibitions (8.7%), and collection quality (8.1%).

The top negative themes from negative and mixed comments for student-targeted exhibitions were overcrowding (9.8%), limited number of guides (6.2%) and exhibition navigation (5.9%).

5.6. National Programmes – Event Sentiment & Feedback dashboard

The “National Programme” dashboard facilitates exploration of museum feedback - visit scores and written sentiment – for multiple V&A board roles to support strategic aims 1 and 4 (i.e., rebuild and broaden audiences; further curatorial practices for modern audiences). The interactive design encourages utility by the board and secondary users, with slicers enabling drilldown to discover underlying patterns in visitor type behaviour or feedback, thus enabling consideration of whether programming connects with the museum’s target audiences.

The dashboard presents the % of exhibition visitors who provided written feedback; and breaks this down by diversity group to understand gender and age representation, and potential self-selection bias. The top 5 positive and negative themes from written sentiment are presented alongside national visit feedback score, and a bar chart dissecting experience sentiment (positive, mixed, and negative) per age group.

Mission 1 and 4: As a whole, written sentiment covered 1 in 20 (5.4%) museum event visitors and was submitted on average 3 days after a museum visit. Feedback had relatively even coverage of all age groups and gender (where known/provided) and was predominantly positive across all age groups (see figure 30).

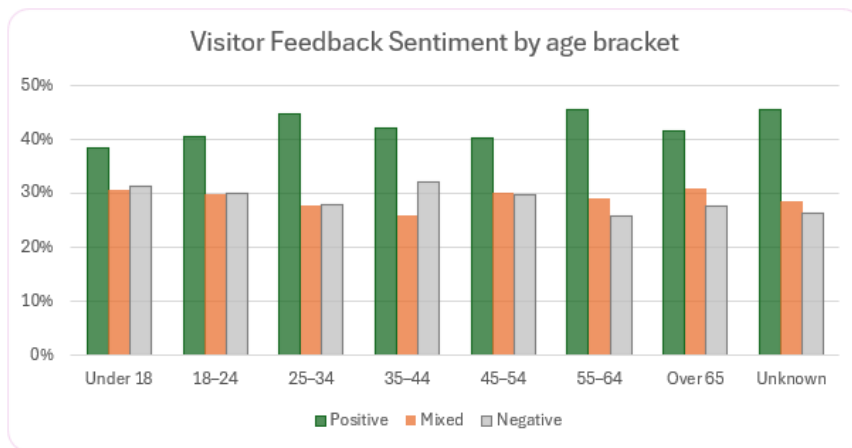


Figure 30: Visitor feedback sentiment by age

Mission 1 and 4: Consideration of broad themes of all written feedback is helpful. Table 2 provides the % breakdown of positive and negative feedback comments related to each broad theme. Although ‘on-site visitor experience’ aligned to the greatest positive theme, the % of negative comments attributed is greater encouraging further exploration. Sentiment towards ‘Events and Programming’ is vastly positive; whilst ‘Policies and Accessibility’ markedly negative. Drill-down into narrow themes associates this to camera-related policy for onsite visitors.

Broad Theme	Positive Broad Theme %	Negative Broad Theme %
On-site Visitor Experience	23.9%	35.6%
Digital Experience	23.7%	23.5%
Events & Programming	21.4%	3.9%
Collections & Archive	13.4%	7.0%
Facilities	8.2%	5.4%
Staff & Customer Service	8.1%	5.3%
Policies & Accessibility	1.2%	14.3%
Pricing	0.0%	5.0%

Table 2: breakdown of positive and negative feedback

Mission 1 and 4: Across all feedback, and drilling through to narrow themes, the top positive themes were: exhibition quality (21.2%), collection quality (8.4%), and interactivity of exhibitions (8.7%). Top drivers of negative feedback were tied to onsite issues: exhibition overcrowding (9.9%), camera policy (5.8%), queues (5.6%), and an overcrowded shop (5.4%). Nationally satisfaction scores (provided per museum visit and attributed to 10k visits, 57k visitors) indicate fair satisfaction over time when averaged; albeit virtual visitors fluctuate more sharply in satisfaction every few months (coinciding with event programming) (see figure 31).

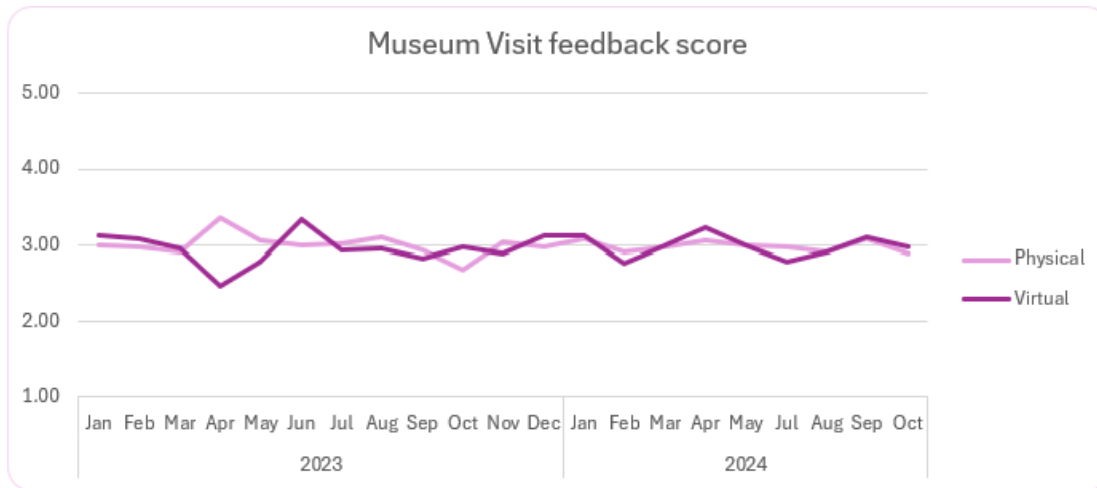


Figure 31: Museum visit feedback score

Mission 1 and 4: Focusing on the V&A’s targeted growth area – family-targeted events - 5.2% of visitors appear to have shared written comments. Positive themed feedback was broadly received from all age groups, particularly those aged 25-34 and 55-64. However, a greater distribution of mixed and negative comments is seen (see figure 32).

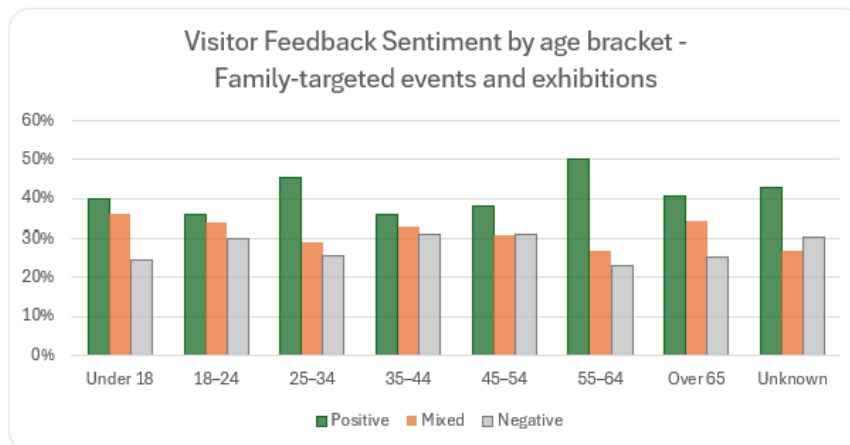


Figure 32: Age based sentiment feedback

Top positive and negative themes for family-targeted events and exhibitions broadly mirrored national findings, however museum heating (7.6% of comments) and items on loan (6.5%) were identified narrow-theme issues. Museum visit scores for family targeted events are marginally higher than all programming, with virtual-visitor average feedback scores showcasing a positive trajectory from July-2024 (see figure 33). A spike of positive sentiment was also experienced in April, linked principally to the last month of “Lost and Found Yokai” (different to scores in, February to March) and the workshop for the Fashion Show.



Figure 33: Family event feedback trends

Isolating feedback to virtual-visitors of family-programmed events, ‘video’ ‘catalogue’ and ‘interactive’ quality were praised; however, video / site accessibility and loading speed were criticised. The largest responding group providing this feedback were 18–24-year-olds (16%) who predominantly provided mixed feedback (66%) (see figure 34).

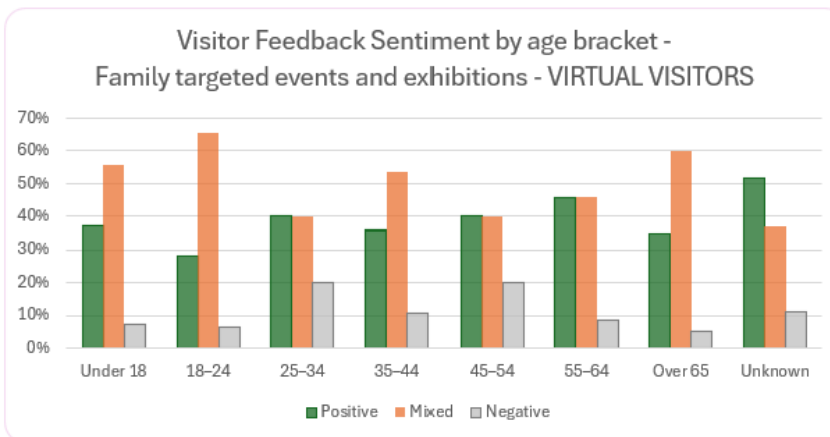


Figure 34: Virtual visitor sentiment by age

Physical visitors to family-targeted events had relatively high proportion of solely-negative comments, e.g., 18-24 years olds (41%), 35-44 years old (47%). However, under 18s were predominantly positive, as were 55 years olds (see figure 35). Top negative themes associated to overcrowding (17.3%), museum heating (11.5%) and items on loan (9.9%).

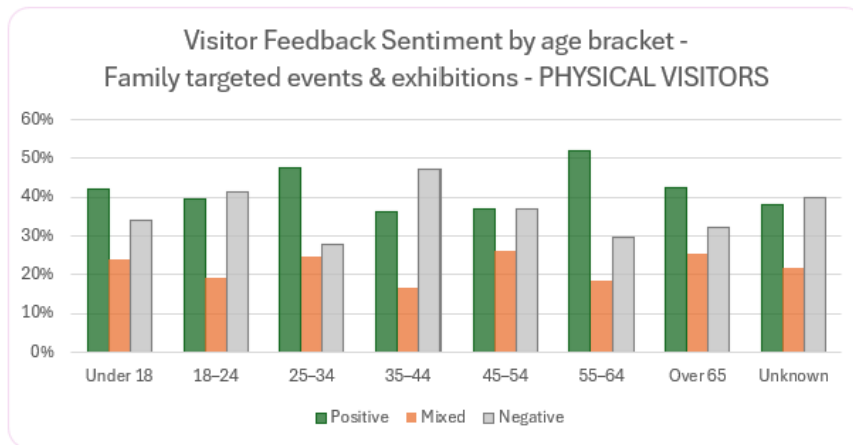


Figure 35: Physical visitor sentiment by age

From this analysis, the hypothesis “*Diversified programming in 2024, towards student and family-targeted exhibitions and events (over adult programming), has grown museum sentiment, engagement and visitor numbers*” is considered partially true.

Attendance of family-targeted events is 41% lower in 2024 based on average attendees per exhibition. National programmes feedback scores evidence gradual inclining sentiment, bar in the Young V&A museum which scores are middling across 2023 and 2024 and projected to decline between October – December 2024. Whilst physical-visitor satisfaction for student and family-targeted exhibitions has received positive sentiment (e.g., by 18–24-year-olds); virtual visitors express mixed experience praising exhibition quality but criticising digital experience, with feedback scores undulating per event.

6. Ethical and Contextual Considerations and limitations

When preparing the V&A Museum’s “donations” dataset for analyses, GDPR’s requirements guided the cleaning process. The dataset originally contained direct identifiers such as donor names, email addresses and full postal addresses. In accordance with the principles of data minimisation and purpose limitation, unnecessary identifiers such as names, house numbers, street names and post codes were removed as they were not required for the analytical objectives (ICO, 2026). Weitzenboeck et al. (2022) reinforce this approach by stating that identifiers significantly increase re-identification risk, making their removal essential when datasets are repurposed for analytics.

Some fields were retained where a legitimate analytical purpose existed. Regional information (e.g., Liverpool, London, Glasgow) was kept to enable geographical donation analyses, whilst email addresses were retained to identify potential repeat donors. This reflects Stalla-Bourdillon’s (2025) argument that identifiability must be assessed in context and that data can be lawfully retained when the purpose is compatible, balanced and appropriately safeguarded. The European Data Protection Board similarly emphasises embedding privacy into all processing operations and ensuring that further processing does not undermine individuals’ rights to privacy (EDPB, 2026). Although email addresses were not ultimately used in the final analyses, their limited and controlled retention reflected a balanced approach to potential analytical value.

By removing unnecessary identifiers and restricting analysis to summarised, non-identifiable elements, the BI workflow reduced the risk of re-identification and met GDPR expectations for ethical, transparent and accountable data handling. Furthermore, all analysis was conducted at

a totalled level (e.g., total donations by event), ensuring that no individual donor could be identified within the analytical outputs.

The Events dataset contains no personal identifiers and is fully combined, which reduces privacy risks (Chevrier et al., 2019); however, ethical considerations still arise in how the data is interpreted across the wider dashboards. The dataset represents only sixteen events over a limited two-year period, indicating its scope is narrow and cannot be generalised to the V&A's full exhibition programme, meaning conclusions must be framed cautiously to avoid overstating trends in attendance, revenue, or audience engagement (Shaheen et al., 2023). The target audience categories (Students, Adults, Families) are broad and do not reflect demographic diversity, so insights must avoid implying behavioural traits or value judgements about real visitor groups. Data cleaning is essential to ensure responsible reporting, particularly where inconsistencies such as negative attendance values, duplicated events, or incorrect revenue calculations could distort financial or operational insights used in the dashboards (Hayes, 2004). As ticket revenue alone does not determine an event's success, findings from this dataset must be contextualised alongside the other four datasets to avoid misrepresenting performance or overlooking cultural, educational, or strategic value. Overall, ethical use of this dataset requires transparency about its limitations, careful framing of insights, and recognition that it provides only one perspective within the museum's broader operational picture (Callaghan, 2020).

The feedback dataset contained written sentiment, alongside visit date, visitor type, submission date, gender and age group. Personal data risk was assessed as low linking to previous pseudonymisation (i.e. adoption of Feedback id) and use of banded age categories; albeit use of individual comments as written could prove identifiable for the individual. Theming and aggregating the sentiment data thus reduces legal and financial risk associated to using this data.

No contextual information was provided on feedback collection method used by the V&A. How surveys are designed, targeted, completed and analysed are important when determining the trust, reliability and relevance of feedback themes and assessing the potential for bias (Bolton and Brace, 2022). Thus, validation of analytical results is recommended as is understanding of the feedback score rating scale for the museum visit dataset.

For V&A board understanding of potential partiality of sentiment the National Programmes Dashboard features a radar chart to guide where returned feedback is more informed by select age and/ or gender cohorts.

With sentiment feedback representing 5.4% of exhibition visitors, the potential of non-response bias is raised as an ethical consideration as the results may not reflect the views of the whole population (Smironva et al., 2020). Undercoverage bias is also a risk if the survey is conducted via online means only. Understanding these risks is particularly relevant to reduce the organisational and societal impact from enacting incorrect decisions based on inferred conclusions or untriangulated themes. Triangulation with independent sources, e.g., Trip Advisor reviews could build confidence and validity of feedback themes albeit due process would be required and consciousness that these reviews could incur self-selection bias (i.e. where the person who organises the museum visit finds the activity more favourable, and is more publicly favourable in their feedback) and come from existing engaged audiences (Smironva et al., 2020).

The preparation of the gift shop sales dataset was completed in line with GDPR principles, particularly data minimisation, purpose limitation, and accountability. The raw dataset contained transaction-level records, including timestamps, product details, pricing, and unique transaction identifiers. Although not directly identifiable, such data presented a risk of indirect re-identification, especially when combined with other datasets. Identifiers and unnecessary variables were either removed or generalised during cleaning to mitigate this risk, consistent

with evidence that transactional data could enable behavioural tracing when retained at high resolution (Rocher et al., 2019; Cohen et al., 2023).

The cleaned dataset retained only variables with clear analytical value, such as product categories, aggregated revenue, and temporal groupings (e.g., monthly sales). Highly specific timestamps were reduced to broader intervals to prevent inference of individual purchasing behaviour, reflecting contemporary data-minimisation expectations that emphasise limiting granularity unless analytically justified. This approach represented a proportionate balance between analytical utility and privacy, aligning with the principle that identifiability must be assessed contextually and that data should be retained only where necessary and justified (Dwork and Roth 2021). All outputs were aggregated to prevent isolation of individual transactions or behavioural patterns.

The dataset was contextually limited, reflecting sales within a single retail environment over a specific period. This restricted generalisability beyond the gift shop and analysed timeframe. It did not account for external factors such as seasonality, exhibition programming, tourism patterns, or economic conditions; therefore, observed trends should not be interpreted as causal. Overgeneralisation could misrepresent performance and undermine decision-making (Shaheen et al., 2023).

Data quality within the raw dataset posed further ethical considerations. Issues including missing values, duplicate transactions, and anomalous pricing required cleaning and validation. While necessary, these processes involved subjective judgement (e.g., deletion vs imputation), which may introduce bias. Recent research highlights that inappropriate handling of missing data can distort statistical relationships and reduce analytical validity, reinforcing the need for transparent and methodologically justified decisions (Van Buuren, 2018; Jakobsen et al., 2017).

A significant limitation was the lack of demographic data, which reduced privacy risk but restricted the assessment of inclusivity, equity, and purchasing behaviour across different groups. This required avoiding unsupported assumptions about customer characteristics or preferences. Reliance on transactional data alone also limited performance evaluation, as revenue does not capture the cultural, educational, or experiential value of the museum retail offering. Ethical analysis therefore required contextualisation with complementary datasets (e.g., visitor engagement or feedback) to avoid purely financial interpretations (Callaghan, 2020).

7. Recommendations for V&A Board

7.1. Director, Deputy Director & Chief Operating Officer

- Diversify visitor flow: Reduce reliance on flagship exhibitions by strengthening mid-tier and smaller programmes to stabilise attendance.
- Manage capacity effectively: Introduce timed entry, real-time monitoring, and improved signage to address overcrowding and queues.
- Standardise experience quality: Implement consistent operational standards across all event types, particularly workshops.
- Improve data capture: Reduce “Unknown” categories through better ticketing, POS, and CRM integration to support decision-making.
- Enhance digital experience: Invest in website performance and accessibility to improve virtual visitor satisfaction.

- Act on feedback faster: Use real-time feedback loops to address recurring issues (e.g., overcrowding, facilities).
- Align resources with demand: Adjust staffing and operations to seasonal trends to improve efficiency.
- Broaden audience engagement: Adapt operations to better attract families and adult visitors, not just students.

7.2. Programme Development

- Attention should be given to the means of getting data, this is due to huge portions of donations coming from unidentified sources. An audit of the donation process is recommended.
- Implement a year-round donation programme to fill the gap in between the early and later months, perhaps season themed campaigns tied to events.
- Adopt a tracking system for event donations as 80% of donated sums can't be attributed to any specific event.
- Conduct deeper level survey on the causes of negative sentiment, a qualitative research to uncover the driver(s) of visitor dissatisfaction.
- Invest more resources on the Cartier event, as it was the highest grossing event.

7.3. Commercial Director

- Maintain a balanced exhibition programme, as visit activity is evenly distributed and no single exhibition dominates engagement
- Optimise exhibition scheduling and promotion during lower-demand months, as visit frequency fluctuates despite continuous programme coverage
- Improve consistency in visitor experience, addressing the high proportion of neutral and negative sentiment across exhibitions
- Expand interactive events such as workshops and installations, which show marginally higher engagement and encourage longer visit durations
- Continue strengthening digital engagement, as virtual visits generate comparable revenue per visit to physical visits
- Improve the recording of visitor type and event, particularly where “Unknown” categories appear, as these limit the accuracy of cross-dashboard analysis.
- Focus retail strategy on high performing product categories such as homeware, stationery and accessories to maximise commercial revenue

7.4. Director of Finance & Resources

- Strengthen retail and ticket-based income to improve financial resilience.
- Targeted programming, marketing, and retail initiatives in spring, summer, and autumn to support more stable annual income.
- Designing targeted offers for adult and family-based events to encourage unrepresented audiences to physically attend.
- Expand and replicate high-performing event themes to enhance commercial returns.
- Optimise micro-donation mechanisms for physical visitors and digital prompts for virtual audiences to provide stable income.

- Prioritise value-driven programming and pricing, rather than attendance growth as a revenue strategy.

7.5. Learning, National Programmes and Young V&A

- Monitor average attendance for family-focused events (KPI) to understand growth of audience over time. If average attendance continues to be 41% lower potential review of exhibition programming or pricing may be required to maintain audience numbers.
- Investigate further the Young V&A feedback scores, which indicated potential disconnect between programmed exhibitions and the museum audience. Consider top sentiment findings to understand if overcrowding (top negative theme for family targeted events, 17.3%) and items being on loan (9.9%) are reducing exhibition experience.
- Increase programming of workshop events, particularly targeted to virtual visitors, to encourage creative confidence and increase experience value and engagement scores.
- Triangulate thematic analysis findings for student targeted events to understand if these exhibitions require content re-shaping to engage younger audiences, or to fix digital issues limiting cohort engagement.

7.6. National Programming Visitor Sentiment

- Investigate if website improvements are required to support enhancement and growth of the virtual visitor experience, particularly with family targeted programming.
- Commission a behavioural survey, or introduce KPIs to analyse and assess how virtual audiences interact with V&A digital exhibitions, and spend time on the website. Aligned with Le Dinh et al. (2024), insight on customer journey could target website enhancements on key touchpoints to invest in and improve visitor experience.
- Investigate negative themes identified from sentiment to quantify / validate issues and target remedial action (e.g., overcrowding, queues for exhibitions, camera policy etc.)

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9. Appendices

Appendix	Document
A	Data Dictionary
B	Thematic categories for feedback comments
C	Strategic questions datasets could answer

9.1. Appendix A: Data Dictionary

9.1.1. Events & Exhibitions

Field Name	Description	Data Type/Format	Values/Domain
Event ID	Unique identifier for each event.	Number	Positive integers
Event Name	Official name or title of the event.	Text	Free text
Start Date	The first day of the event.	Date (dd/mm/yyyy)	Valid excel date values
End Date	The last day of the event.	Date (dd/mm/yyyy)	Valid excel date values
Type	Category of event.	Text	Exhibition, Workshop, Installation, Festival, Course
Target Audience	Intended primary audience.	Text	Student, Adults, Families
Attendance Number	Count of attendees for the event.	Number	Positive integers
Ticket Price	Price per ticket in GBP.	Number (Currency)	>0
Revenue Generated (£)	Total revenue generated from the event (Attendance * Ticket Price).	Number (Currency)	>0

9.1.2. Museum Visits

Field Name	Description	Data Type/Format	Values/Domain
Date	Date of visit. (A required field)	Date (dd/mm/yyyy)	Valid excel dates value
Type	Type of visitor.	Text	Physical, Virtual, Unknown
Visitor Group Size	The number of people visited in one group.	Number	Positive integers

Duration (Minutes)	The length of time the visitor was there in minutes.	Number	Positive integers
Event Name	Official name or title of the event.	Text	Free text
Region	The location of the museum.	Text	Online, Dundee, East Storehouse, South Kensington, Young V&A, Unknown
Feedback Rating	Rating of the visit on a scale of 1 to 5.	Number	1, 2, 3, 4, 5

9.1.3. Gift Shop Sales

Field Name	Description	Data Type/Format	Values/Domain
Sale ID	Unique identifier for each sale.	Number	Positive integers
Visitor Type	Type of visitor.	Text	Physical, Virtual, Unknown
Date	Date of sale.	Date (dd/mm/yyyy)	Valid excel dates value
Product Name	The name of the product sold.	Text	Free text
Quantity Sold	The number of items sold.	Number	Positive integers
Unit Price (£)	The price of the product.	Number (Currency)	>0
Total Sale Value (£)	The total sale amount (Quantity * Unit Price).	Number (Currency)	>0
Associated Event	Official name or title of the event.	Text	Free text
Category	The category of the product sold.	Text	Accessories, Apparel, Art, Books, Games,

			Homeware, Jewellery, Stationary, Unknown
Payment Type	The payment method used.	Text	Card, Cash, Unknown
Donation round up	Whether the payment was rounded up for donation.	Text	Yes, No
Donation (£)	The amount of donation paid.	Number (Currency)	>0

9.1.4. Donations

Field Name	Description	Data Type/Format	Values/Domain
Donation ID	Unique identifier for each donation.	Number	Positive integers
Date	Date of donation.	Date (dd/mm/yyyy)	Valid excel dates value
Type	Type of donation made.	Text	Online, Physical, Unknown
Donation Amount	The amount of donation paid.	Number (Currency)	>0
Event Name	Official name or title of the event.	Text	Free text
Museum	The location of the museum.	Text	Online, South Kensington, V&A Dundee, V&A East Storehouse, Young V&A
Benefactor Type	The type of people involved in the donation.	Text	Corporation, Family, Individual, Unknown
Contact Email	The email of the person who made the donation.	Text	Free text

Town	The hometown of the person who made the donation.	Text	Free text
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9.1.5. Feedback

Field Name	Description	Data Type/Format	Values/Domain
Feedback ID	Unique identifier.	Number	Positive integers
Date of Visit	Date of visit.	Date (dd/mm/yyyy)	Valid excel dates value
Visitor Type	Type of visitor.	Text	Physical, Virtual, Unknown
Event Name	Official name or title of the event.	Text	Free text
Feedback	The feedback comments provided by the visitor.	Text	Free text
Feedback Theme	The overall theme of the feedback.	Text	Positive, Mixed, Negative
Positive Theme	The specific theme of the positive feedback.	Text	Free text
Positive Broad Theme	The broad theme of the positive feedback.	Text	Collections & Archive, Digital Experience, Events & Programming, Facilities, On-site Visitor Experience, Policies & Accessibility, Staff & Customer Service
Negative Theme	The specific theme of the negative feedback.	Text	Free text

Negative Broad Theme	The broad theme of the negative feedback.	Text	Collections & Archive, Digital Experience, Events & Programming, Facilities, On-site Visitor Experience, Policies & Accessibility, Staff & Customer Service
Submission Date	The date the feedback was submitted.	Date (dd/mm/yyyy)	Valid excel dates value
Gender	Gender of the person who provided feedback.	Text	Female, Male, Prefer not to say, Unknown
Age	Age band of the person who provided feedback.	Text	Under 18, 18-24, 25-34, 35-44, 45-54, 55-64, Over 65, Unknown

9.2. Appendix B: Thematic categories for feedback comments

Broad Theme	Narrow Theme
Collections & Archive	Archive Quality
	Catalogue quality
	Collection Quality
	Items on loan
	Negative experience
	Overwhelming archive experience
	Unenjoyable experience
Digital Experience	Broken links
	Ease of online collection search
	Exhibition Quality
	Immersive Tour
	Insightful webinar
	Interactive Quiz
	Interactive timeline feature
	Lack of interactive features
	Loading speed
	Quality videos
	Site Navigation / Search options
	Tour Presentation
	Video Quality

Broad Theme	Narrow Themes
Events & Programming	Enjoyable
	Exhibition Quality
	Short event timeframe
Facilities	Video duration too short
	Café
	Shop overcrowding
	Shop Quality
On-site Visitor Experience	
	App guide useability
	Downloadable Guides
	Exhibition Navigation
	Exhibition Presentation
	Interactive exhibits
	Interactive museum map
	Live demonstrations
	Museum heating
	Overcrowded
	Poor lighting
	Poor signage
	Queues
	Camera policy

	Virtual tour experience
	Website appearance
	Website design
	Website Feature - exhibit zoom
	Website Navigation
	Website not intuitive
	Website personal gallery
	Website search errors
	Website speed
	Website Visuals

Policies & Accessibility	Venue accessibility
	Video accessibility
	Virtual accessibility options
Pricing	Café prices
	Ticket Prices
Staff & Customer Service	Helpful Friendly Staff
	Limited number of guides
	Passionate Staff

9.3. Appendix C: Strategic questions datasets could answer

9.3.1. Strategic Mission 1 – Rebuild Visitor Numbers

Role	Dataset	Question	Insight
Director/COO Director of Learning, NP&Y	Exhibitions & Events	Which event types attract the most visitors?	Identifies high demand programming
Director of Learning, NP&Y	Exhibitions & Events	Which audiences are under-represented?	Shows where outreach is needed
Director/COO, Finance & Resources	Museum Visits	How many visits occur per month?	Tracks visitor flow
Director/COO, Director of Learning, NP&Y	Museum Visits	Are physical or virtual visits growing?	Channel engagement trends
Director of Learning, NP&Y	Feedback	Which visitor types give the most positive feedback?	Experience quality by audience
Director of Learning, NP&Y	Feedback	Are digital events altering/ balancing the V&A audience profile?	Audience profile change
Commercial, Finance/Resource	Gift Shop Sales	Do physical visitors spend more than virtual visitors?	Behaviour differences
Development, Finance/Resource	Donations	Which regions donate most frequently?	Geographic engagement
Director of Learning, NP&Y	Events + Museum Visits	Do high attendance events also generate more visits and longer dwell times?	True engagement, not just footfall
Commercial	Events + Gift Shop	Which events convert visitors into spenders?	Event driven commercial impact
Director of Learning, NP&Y. Director/COO	Events + Feedback	Do popular events also have high satisfaction?	Quality vs quantity
COO/Finance	Museum Visits + Donations	Do visitors from certain regions donate more?	Regional engagement strength

Director of Learning, NP&Y	Feedback + Museum Visits	Do satisfied visitors stay longer?	Experience depth
Commercial	Gift Shop + Feedback	Do happy visitors spend more?	Satisfaction → spending behaviour

9.3.2. Strategic Mission 2 – Transform Access & Creative Relevance

Role	Dataset	Question	Insight
Director of Learning, NP&Y	Exhibitions & Events	Are events showcasing diversity?	Diversity in events
Director/COO	Exhibition & Events + Museum Visits	Are new sites building visitor numbers and widening access?	Launch of V&A East Storehouse
COO, Director of Learning, NP&Y	Feedback	Which themes dominate positive feedback?	What resonates creatively
Director of Learning, NP&Y	Feedback	Which events receive negative comments about access?	Accessibility issues
Commercial	Museum Visits	Which events have the longest dwell times?	Depth of engagement
Commercial	Gift Shop Sales	Do products linked to diverse cultural themes sell well?	Diversity in retail
Development	Donations	Which events inspire giving?	Emotional resonance
Director of Learning, NP&Y	Events + Feedback	Which events are most diverse and creatively impactful?	Creativity & diversity validated by sentiment
Commercial	Events + Museum Visits	Do creative skills events lead to longer visits?	Engagement depth
Commercial	Events + Gift Shop	Do creative events drive sales of books/kits?	Creative confidence indicators
Director of Learning, NP&Y	Feedback + Museum Visits	Do visitors who stay longer report more positive themes?	Engagement → satisfaction

Development	Events + Donations	Which events inspire donations due to creative impact?	Creativity → giving
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9.3.3. Strategic Mission 3 – Inspire Creative Confidence & Skills

Role	Dataset	Question	Insight
Director of Learning, NP&Y	Exhibitions & Events	Which events target young people or learners?	Youth engagement
Director of Learning, NP&Y	Feedback	Do younger visitors report higher satisfaction?	Creative confidence
Director of Learning, NP&Y	Feedback	Which positive themes relate to learning or inspiration?	Creative uplift
Director of Learning, NP&Y	Feedback	Is sentiment shifting in different age groups?	Youth & Under 25-year-old engagement
Commercial, Finance	Gift Shop Sales	Do learning materials sell well?	Creative skill development
Director of Learning, NP&Y	Museum Visits	Do school or family groups stay longer?	Engagement with learning
Development	Donations	Do education focused events attract donors?	Support for creative skills
Director of Learning, NP&Y, Commercial	Events + Feedback	Which creative skills events generate the best sentiment?	Inspirational programming
Director of Learning, NP&Y	Events + Museum Visits	Do creative events attract larger or younger groups?	Youth engagement
Commercial	Feedback + Gift Shop	Do inspired visitors buy creative kits/books?	Creative confidence → spending
Director of Learning, NP&Y	Feedback + Museum Visits	Do younger visitors stay longer or rate higher?	Youth experience quality
Development	Events + Donations	Do creative skills events inspire giving?	Creative confidence → philanthropy

9.3.4. Strategic Mission 4 – Encourage Effectiveness, Sustainability & Inclusion

Role	Dataset	Question	Insight
COO, Director of Learning, NP&Y	Feedback	Are certain demographics less satisfied?	Inclusion gaps
Director of Learning, NP&Y	Feedback	What negative themes relate to accessibility?	Barriers to access
Director of Learning, NP&Y	Museum Visits	Which regions are under-represented?	Inclusion gaps
COO	Exhibitions & Events	Which audiences are not attending?	Outreach needs
Development	Gift Shop Sales	Are donation round-ups common?	Community support
Development	Donations	Which benefactor types are most engaged?	Sustainable funding
Development	Donations	How frequently do benefactors donate again? (donator retention). (How much do benefactors typically donate in subsequent donations?)	Sustainable funding Lifetime value of consumers
Director of Learning, NP&Y	Feedback + Museum Visits	Do accessibility complaints correlate with shorter visits?	Barriers to engagement
Director of Learning, NP&Y	Events + Feedback	Which events fail to engage diverse audiences?	Inclusion issues
Development	Museum Visits + Donations	Do local visitors donate more or less?	Community support
Development	Gift Shop + Feedback	Are certain groups less likely to donate or round-up?	Inclusion in giving
Director of Learning, NP&Y	Events + Museum Visits	Are certain events failing to attract diverse regions?	Geographic inclusion

9.3.5.Strategic Mission 5 – Build Financial Resilience

Role	Dataset	Question	Insight
Finance, COO	Exhibitions & Events	Which events generate the most revenue?	Revenue drivers
Commercial	Gift Shop Sales	Which categories/ products are most profitable?	Retail performance
Development, COO	Donations	Which donation types bring in the most money?	Fundraising strength
Development	Museum Visits	Which regions produce high value visitors?	Geographic revenue optimisation
Commercial	Feedback + Gift Shop	Does satisfaction correlate with spending?	Experience → revenue
Finance, Commercial, COO	Events + Gift Shop + Donations	Which events generate the highest combined revenue?	True event ROI
Commercial	Feedback + Gift Shop	Do satisfied visitors spend more?	Satisfaction → commercial impact
Commercial	Museum Visits + Gift Shop	Which visitor types spend more per visit?	Behavioural revenue insights
Development	Events + Donations	Which events inspire giving?	Emotional ROI
Development	Museum Visits + Donations	Which regions produce the highest donor value?	Geographic fundraising strategy